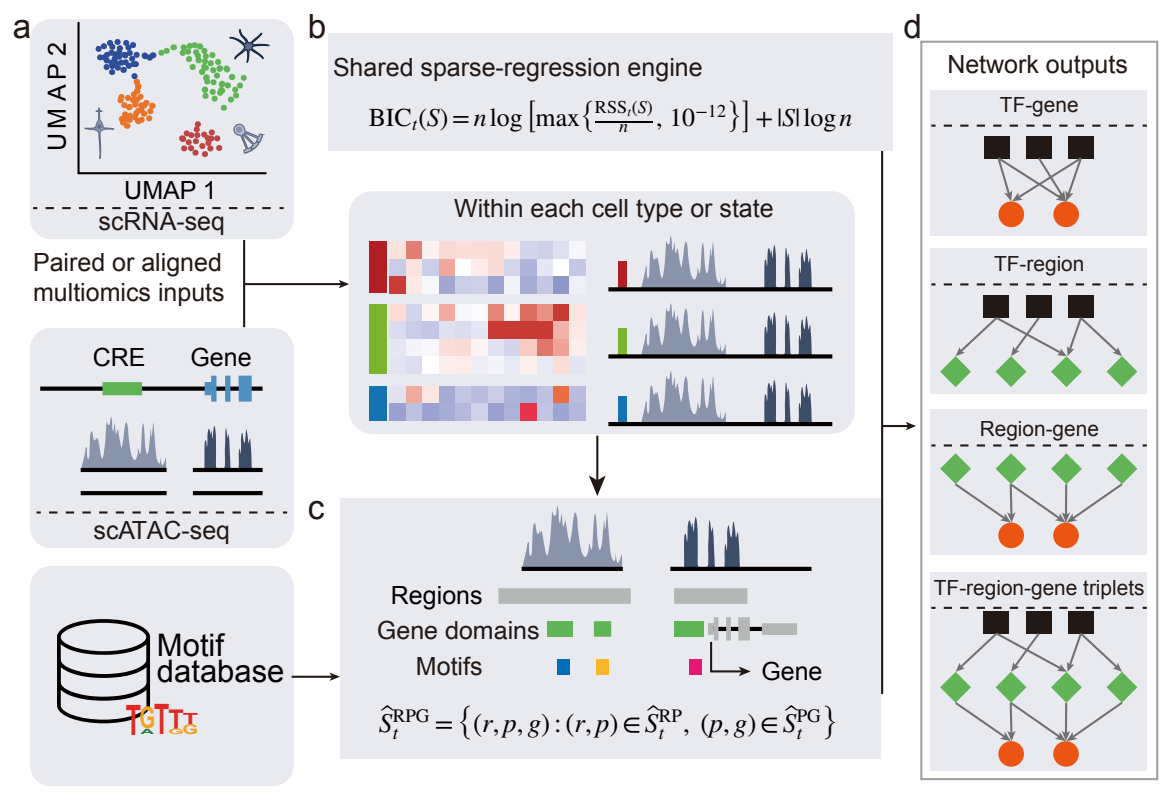

a
UMAP 2
UMAP 1
scRNA-seq
Paired or aligned multiomics inputs
CRE
Gene
scATAC-seq
Motif database
b
Shared sparse-regression engine
$\mathrm{BIC}_t(S) = n \log\left[\max\left\{\frac{\mathrm{RSS}_t(S)}{n}, 10^{-12}\right\}\right] + |S| \log n$
Within each cell type or state
c
Regions
Gene domains
Motifs
Gene
$\hat{S}_t^{\mathrm{RPG}} = \left\{(r,p,g) : (r,p) \in \hat{S}_t^{\mathrm{RP}},\ (p,g) \in \hat{S}_t^{\mathrm{PG}}\right\}$
d
Network outputs
TF-gene
TF-region
Region-gene
TF-region-gene triplets